Math 526, F09
Quiz 7

Name: Answer
SSN: xxx-xx-

<u>**Instructions:**</u> **There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 6 & 4 & 3 \\ -2 & -6 & 0 & -1 \end{bmatrix}$.

(a) Transform $\mathbf{A}$ into the reduced echelon form $\mathbf{R}$.

$$A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 6 & 4 & 3 \\ -2 & -6 & 0 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$

$$\longrightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 3 & 0 & \frac{1}{2} \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 3 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(b) Find the dimension and a basis for the column space of $\mathbf{A}$, $C(\mathbf{A})$.

$\dim(C(A)) = 2$, Pivot columns: 1, 3

A basis: $\begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}$

(c) Find the dimension and a basis for the row space of $\mathbf{A}$, $C(\mathbf{A}^T)$.

$\dim(C(A^T)) = 2$, Pivot rows: 1, 2

A basis: $\begin{bmatrix} 1 \\ 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 4 \\ 3 \end{bmatrix}$

(or $\begin{bmatrix} 1 \\ 3 \\ 0 \\ \frac{1}{2} \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ \frac{1}{2} \end{bmatrix}$)

(d) Find the dimension and a basis for the nullspace of $\mathbf{A}$, $N(\mathbf{A})$.

$\dim(N(A)) = 4-2=2$, free variables: x_2, x_4 $\rightarrow$ Solve $AX=0$ $\Leftrightarrow RX=0$

Let $x_2=1$, $x_4=0$, we get $x_1=-3$, $x_3=0$

$$\Rightarrow S_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Let $x_2=0$, $x_4=1$, we get $x_1=-\frac{1}{2}$, $x_3=-\frac{1}{2}$

$$\Rightarrow S_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

A basis for $N(A)$: $\begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 0 \\ -\frac{1}{2} \\ 1 \end{bmatrix}$

(e) What is the dimension of the left nullspace of $\mathbf{A}$, $N(\mathbf{A}^T)$?

$\dim(N(A^T)) = 3-2=1$